

INDUSTRIAL CHEMISTRY

Introduction: Industrial Chemistry is the branch of Chemistry that identifies and solves problems that are met in the exploitation of all raw materials and which tries to produce from it on an appropriate scale, many other products which do not occur naturally, and which are becoming scarce in nature, or whose production methods can be improved upon.

Scope: Industrial chemistry is the science that exists in the exploration of nature's natural resources and so doing serves as a stimulus to the pursuit of the optional course, followed by natural creative evolution, with a view to enhancing the available natural resources.

It is a synthesis of science and technology and hence, enables humanity to experience the benefits of the functions performed by chemistry.

Divisions of Industrial Chemistry.

Industrial Chemistry can be simply classified into two main areas of interrelated activities; viz. Inorganic Industrial Chemistry and organic Industrial Chemistry.

The Inorganic Industrial Chemistry is the application of chemical transformation operations to natural inorganic materials. Activities here include but not restricted to Mineral processing, Extraction of elements, eg Au, Fe, Ag etc.

Large scale production of inorganic substances e.g lime from limestone for cement production, metal manufacture and most fine xcal productions.

Organic Industrial chemistry on the other hand, is the application of chemical transformation processes to natural organic materials.

Activities here include Petroleum Refining and exploration processes, coal mining and or natural gas⁹⁰⁸ extraction and nowadays shale gas extraction. Large-scale production of organic substances or derivatives based on organic substances.

Common examples are: Fermentation processes, herbicides and pesticides manufacture, Production of antibiotics, Soap and detergents, Paints and adhesive manufacture etc.

Chemical Process and Industries

Chemical processes is a combination of all the real reactions involved in the production of a new product be it synthetic or semi-synthetic.

These processes could be refining, synthesis, oxidation, hydration, salt formation and various other methods or operations based on physical phenomena such as: Crystallization, Evaporation, distillation and extraction.

All of the above mentioned processes leads to the transformation of raw materials e.g. Petroleum, cane-sugar, coal, air, limestone & gypsum into finished products such as Petrol, diesel, kerosene, sugar, coke, coal tar, oxygen, cement, fertilizers, Nitric and Sulphuric acids for both man and industrial applications.

While the chemical industry includes industries engaged in the manufacture of chemicals that are both organic and inorganic. For example, Petrochemicals from petroleum refinery, Polymers industries, rubber and plastics from distilleries, Pulp and Paper from Paper industries, Pharmaceuticals from drug industries, Explosives from military and other warfare industries, Others are food additives and flavors from food industries and Confectionaries. etc.

Chemical Process - Technology: This is the complex set of characteristics of the apparatuses in which a chemical process is carried out and which determines its specification.

A chemical process can either be a pilot process, a laboratory process or an industrial process.

An Industrial Chemical process is one in which the operational conditions are vigorous and the apparatuses are constructed from metals or ceramics, with dimensions such that quantities of substances of tons or kilograms per day can be treated or formed.

The assemblage or systematic collection of apparatuses by means of which an industrial chemical process is realised in an industrial chemical plant and the technology used in the setting up of conditions of such a plant, in a special way is what is called chemical Technology.

An industrial chemical plant which is intended to carry out a certain industrial chemical process comprises all the apparatus necessary to satisfy requirements regarding temperature, pressure changes, filtration and drying conditions etc. which may be requested in order to carry out the basic reactions or secondary reactions.

"Basic reactions" are those involving the transformations of prime or primary materials into finished products, that is, into products which the plant was established to produce.

while "Secondary reactions" are those which leads to the formation of side products and means of recovering and or eliminating them.

Major Raw materials and their sources

From the natural environment i.e. the lithosphere, hydrosphere, atmosphere and the biosphere. Industrial chemistry procures raw materials in order to convert them into intermediate products - useful products which are subsequently provided as pure materials to every other kind of industry for processing.

The lithosphere; the bulk of the chemical elements be it metal or non-metals are taken from the earth's crust, and these elements naturally occur in the combined form from which they must be extracted. These include mineral ores, tin ore, lead ore,

limestone, salt or sodium chloride as rock salt in underground deposits and sea water (brine).

Sulphur occurs naturally in some places and is also obtained as a by-product, when sulphide ores of copper or zinc are treated.

Salts forming the basis of many chemical products such as barium and calcium sulphates and fertilizers such as nitrates and phosphates are also derived from the earth's crust. Others are limestone for the cement industry, leams and silica in many varied forms - and silicates for the ceramic and glass industries. Important metals such as iron, Fe, Aluminium Al, copper

Cu are extracted from their respective mineral ores, mostly found in large deposits spread all over the world.

"An ore" is aggregate of minerals from which one or more metals can be profitably extracted.

The non-valuable minerals of an ore are called "gangue". While "minerals" on the other hand are defined as naturally occurring, crystalline, inorganic substances that have definite small range of X-ray composition and physical properties.

Minerals are metalliferous or non-metalliferous. The metalliferous are the ore minerals i.e. ores. While the non-metalliferous minerals are for e.g. salt (NaCl), which occurs as sedimentary rock deposits, Phosphate rocks that occur as marine phosphate sedimentary rocks, limestone, clay (kaolinites) occur as products of residual weathering of some rocks and also as sedimentary deposits.

Other raw materials from the earth crust include: Coal, Petroleum, Natural gas etc.

Hydrosphere — This is where water, brine, ice, brine and hydro power is obtained.

Biosphere — This is where most of essential and other secondary resources such as the celluloses, the plant kingdom. Fats and oils come from plants and animal kingdoms. Wine, enzymes for the biochemical industry are predominantly obtained from the animal kingdom.

The Chemical Process Industries in Nigeria

The following are some of the major Chemical Industries that are found in Nigeria.

1. The Petroleum Industry: This is the major industry and the leading one in the country today.

Its operations are exploration and production of petroleum. Refining and reprocessing operations are also carried out here.

2. Petrochemicals Industry: This is concerned with the production of petrochemicals such as ethene, ethylene and its derivatives. Butane (Butylene), Propane (propylene) etc. Aromatics: Such as Benzene, Toluene, Xylene (C₈H₁₀). Benzene is the starting material for styrene, phenol and a number of fibres and plastics.

3. Fertilizer Industry: This is primarily concerned with the production of various types of fertilizers which help improve soil nutrients and consequently crop yield.

4. Pharmaceutical Industry: This is where production of high quality essential drugs and other health related accessories is carried out. Most of the industries here were jointly owned by foreign and home investors. However, today the country can boast of a large number of these industries that are wholly owned and operated by the country's national.

5. Cosmetics Industry: This is where the much needed cosmetics and other beautician sundries are made. There is a proliferation of these industries in this country today due mainly to its profitability.

6. Plastics Industries: This is where most of ice house hold plastic utensils and other wares are manufactured.

7. Rubber and Footwear Industry: Here materials ranging from fairs, cushions, pillows, shoes are manufactured.

8. Beverage Industries: This where, we bulk of the soft and hard drinks are produced.

9. Pulp and Paper Industry: This is where most paper and other related goods are manufactured.
Most of the goods here used for manufacture of books, news paper and packaging materials.

10. Sugar Industry: This where sugar is produced primarily from cane-sugar; both for immediate consumption and for other industrial purposes.